

Lattice Green’s function of the hyperkagome lattice: modular uniformization at level 30 from an orthogonal differential Galois group

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The lattice Green’s functions (LGFs) of the simple-cubic, body- and face-centred cubic lattices are classical: each is the symmetric square of a second-order (elliptic) operator and admits a closed form in complete elliptic integrals. The three-dimensional *hyperkagome* lattice—the network of corner-sharing triangles realised by the Ir sublattice of the spin-liquid candidate $\text{Na}_4\text{Ir}_3\text{O}_8$ —has resisted such a treatment: Varma and Monien [Phys. Rev. E **87**, 032109 (2013)] reduced its density of states to a threefold integral they “found no way to exactly solve.” We determine its analytic structure and prove that it is *modular*. Working from exact lattice moments, we show that the hyperkagome LGF $G(z)$ (after removing the trivial flat-band pole and exploiting an exact spectral symmetry) is annihilated by a *third-order* Picard–Fuchs operator M of polynomial degree fifteen, which we determine by differential approximants and certify to the guess-and-verify standard of this field—by massive overdetermination (exact annihilation over $\mathbb{Q}$ through order t^{111} : 112 relations against the 57 free coefficients of M) and an independent from-scratch reconstruction (an unconditional creative-telescoping proof that M is the minimal annihilator remains open). The operator M is *irreducible* over $\overline{\mathbb{Q}}(t)$ and is not a literal symmetric square—its Frobenius exponent triples are not the arithmetic progressions that structure would force—yet its symmetric square possesses an explicit *rational* solution: the solution space carries a monodromy-invariant, nondegenerate, symmetric bilinear form, the differential Galois group is the full orthogonal group $\text{O}(3, \mathbb{C})$ with identity component $\text{SO}(3, \mathbb{C}) \cong \text{PSL}(2, \mathbb{C})$, and M is projectively equivalent—through an order-two differential intertwiner, which is exactly why the exponent test does not see it—to the symmetric square of an explicit second-order operator V_2 . Our main result is that V_2 *uniformizes a classical modular curve*: with the level-30 eta quotient $u = [\eta(\tau)\eta(6\tau)\eta(10\tau)\eta(15\tau)/(\eta(2\tau)\eta(3\tau)\eta(5\tau)\eta(30\tau))]^3$, the variable $t = u/(u^2 + 7u + 1)$ is a Hauptmodul of $\Gamma_0(30)^+$ (equivalently $1/t = u + 7 + 1/u$), and the projective normal form of V_2 satisfies the Schwarzian identity $\{\tau, t\} = 2Q_V(t)$ *exactly*—proven, not numerically checked, by combining the Ligozat and Atkin–Lehner structure of u with a pole-degree bound that converts agreement of exact rational q -expansions through order 96 into an identity. The hyperkagome LGF is therefore modular at level $30 = 2 \cdot 3 \cdot 5$ —to our knowledge the first lattice Green’s function realised at a modular level with three distinct prime factors—and the projective monodromy of M is the arithmetic lattice $\Gamma_0(30)^+$, of hyperbolic covolume 3π , with orbifold signature $(0; 2, 2, 2, 2, 2; 1 \text{ cusp})$ derived, not assumed. This proves the closed form whose existence Varma and Monien conjectured. The weight-two period $y_0 = \Phi'/2$ itself is then obtained in explicit closed form, $y_0 = [\rho_0(t)W + \rho_1(t)W']/\sqrt{(1-4t)(1-5t)(1-9t)}$ with $W = qdt/dq$ and explicit rational ρ_0, ρ_1 : a weight-two, *depth-one quasimodular* form on $\Gamma_0(30)^+$ twisted by the determinant character—and the derivative term is provably essential: we show y_0 is not a (meromorphic modular form) $\times$ (algebraic function) at any weight or level, in particular not an eta quotient, so the period is quasimodular rather than modular. The operator remains non-Liouvillian— $G^\circ = \text{SO}(3, \mathbb{C})$ is simple, hence non-solvable—so no algebraic or elementary closed form exists; the closed form is modular and quasimodular (eta quotients and Eisenstein series; a ${}_2F_1$ representation follows in principle via an algebraic pullback), exactly as the orthogonal Galois group demands. We further reduce the LGF, exactly, to a classical two-parameter Watson integral—the body-centred-cubic lattice with first- and second-neighbour hopping at spectral parameter $1 - 6t + 3t^2$ —identifying the object within the classical cubic-lattice Green’s-function literature. We give the operator, its complete Riemann scheme, the invariant quadratic form and its Gram data, the single 3×3 Jordan block at $t = 0$ (maximal log power $n = 2$), the determinant character and its genus-one twist curve $X(\Gamma_0(30)+2, 3, 6)$, the closed-form weight-two period, the Watson reduction, and the exact value $\text{Re } G(1) = 1/9$.

I. INTRODUCTION

The lattice Green’s function (LGF)

$$G(z) = \int_{\text{BZ}} \frac{d^3k}{(2\pi)^3} \frac{1}{z - \varepsilon(\mathbf{k})} = \int \frac{\rho(E) dE}{z - E}, \quad (1)$$

the resolvent of a tight-binding band $\varepsilon(\mathbf{k})$ averaged over the Brillouin zone, is a central object of lattice statis-

tics: it controls random-walk return probabilities, defect and impurity states, and the local density of states $\rho(E) = -\pi^{-1} \text{Im } G(E + i0^+)$ [1, 2]. For the three-dimensional cubic lattices the LGF is a celebrated closed form. Watson’s triple integral for the simple-cubic (SC) lattice [3], and the body- and face-centred analogues, are all expressible through products of complete elliptic integrals K [4–6]; structurally, each of these order-three

operators is the *symmetric square* of a second-order (elliptic) operator [18], which is precisely why an elliptic closed form exists.

Not every lattice is so fortunate. As the local geometry becomes more intricate—decorated, frustrated, or multi-band lattices—the LGF can fail to reduce to elliptic integrals and instead becomes a period of a higher-order or irreducible differential operator (the diamond lattice already signals this trend [2, 6]). The natural three-dimensional frustrated network is the *hyperkagome* lattice: a chiral arrangement of corner-sharing triangles, obtained by removing one quarter of the sites of the pyrochlore lattice in an ordered (space group $P4_132$) fashion. It is realised by the Ir^{4+} moments of $\text{Na}_4\text{Ir}_3\text{O}_8$, a leading three-dimensional quantum-spin-liquid candidate [7–9], which has made its single-particle spectral functions of independent interest.

Varma and Monien [10] computed the hyperkagome density of states and reduced it to a single threefold integral of a rational function of $\cos k_x, \cos k_y, \cos k_z$, remarking that “it seems highly suggestive that a closed form expression can be obtained if a particular complicated yet highly symmetrical integral can be solved,” but that “we have currently found no way to exactly solve” it. That integral has remained open.

We settle its analytic structure and prove its modularity. The hyperkagome LGF is governed by an *irreducible third-order* Picard–Fuchs operator M that is not a *literal* symmetric square, yet whose symmetric square carries a rational solution: the differential Galois group is *orthogonal*, $G = \text{O}(3, \mathbb{C})$ with $G^\circ = \text{SO}(3, \mathbb{C}) \cong \text{PSL}(2, \mathbb{C})$, so M is projectively equivalent to the symmetric square of a second-order operator. The obstruction Varma and Monien met is therefore not a genuine third-order (“ SL_3 -generic”) object with no elliptic reduction; it is a *hidden orthogonal / modular structure*, realised through a nontrivial operator equivalence rather than the function-multiplier and pullback equivalences one usually tests. We obtain the operator explicitly, certify this structure by exact computation, and thereby explain *why* a naive symmetric-square reduction was never found. We then prove the closed-form structure outright: the second-order operator underlying M is the uniformizing operator of the genus-zero modular curve $X(\Gamma_0(30)^+)$, the natural variable t is a Hauptmodul given by an explicit level-30 eta quotient, and the hyperkagome LGF is modular at level $30 = 2 \cdot 3 \cdot 5$ —to our knowledge the first three-dimensional LGF known to require such an equivalence, and the first lattice Green’s function realised at a modular level with three distinct prime factors.

II. HYPERKAGOME LATTICE AND ITS GREEN’S FUNCTION

The hyperkagome lattice has twelve sites per cubic cell, each with coordination number four; its nearest-neighbour tight-binding Bloch Hamiltonian $H(\mathbf{k})$ is a

12×12 Hermitian matrix. We fix the hopping $t = 1$ and the energy origin at the band centre. Diagonalising $H(\mathbf{k})$ over the Brillouin zone yields the spectrum shown in Fig. 1. Three exact features organise everything that follows.

(i) *A flat band, and the line-graph structure.* The hyperkagome lattice is the line graph of a three-regular “premedial” net Γ (eight vertices per cell, twelve edges): its sites are the edges of Γ and its triangles are the vertices of Γ . For the line graph $L(\Gamma)$ of a d -regular graph one has $A(L(\Gamma)) = B^\top B - 2I$ with B the incidence matrix, so the spectrum is $\{\mu_i + d - 2\}$ together with the eigenvalue -2 of multiplicity (edges) – (vertices). Here $d = 3$, giving four flat bands at $E = -2$ (spectral weight exactly $1/3$, i.e. $\frac{12-8}{12}$); this contributes a pole $\frac{1}{3}(z+2)^{-1}$ to $G(z)$ and is the flat-band pole written $1/(t_{\text{VM}}+1)$ in Ref. [10], whose energy variable is $t_{\text{VM}} = E/2$ (the correspondence, including this pole and its weight, is reproduced in their variable in the Supplemental Material). The eight dispersive bands are $E = \mu_i + 1$ and fill $[-2, 4]$, since the adjacency eigenvalues μ_i of a three-regular graph lie in $[-3, 3]$.

(ii) *An exact reflection symmetry, from bipartiteness.* A graph has an adjacency spectrum symmetric about 0 if and only if it is bipartite; we verify that the premedial net Γ is bipartite (it admits a translation-consistent two-colouring, checked on finite blocks up to 5^3 cells, and its cycles are of even length). Through the line-graph shift $E = \mu + 1$ this makes the dispersive density of states *exactly* symmetric about $E = 1$; independently, all odd central moments $\int \rho_{\text{disp}}(E)(E-1)^{2m+1}dE$ vanish in exact arithmetic to order 239. Hence the dispersive part of G is odd in $\zeta \equiv z - 1$, and is a function of $t \equiv \zeta^{-2}$.

(iii) *Integer moments.* The per-site spectral moments $m_n = \frac{1}{12} \text{Tr} H^n$ are exact integers, $m_{0..10} = 1, 0, 4, 4, 28, 60, 260, 756, 2828, 9292, 33384$; we computed them to m_{230} by closed-walk enumeration. Among the ordered quarter-depletions of the pyrochlore lattice, the chiral hyperkagome enantiomorphs are the ones reproducing this sequence (hexagon-containing depletions already differ at $m_6 = 264 \neq 260$); this selection is checked separately and is not needed for what follows.

III. THE PICARD–FUCHS OPERATOR

Because $G(z)$ is the diagonal of a rational function over the torus, it is holonomic (D-finite): it satisfies a linear differential equation with polynomial coefficients [11]. We determine that equation from the moments. Removing the flat band, $d_n = m_n - \frac{1}{3}(-2)^n$, and forming the even central moments $\nu_m = \int \rho_{\text{disp}}(E-1)^{2m}dE$, the generating function $\Phi(t) = \sum_m \nu_m t^m$ (with $t = \zeta^{-2}$) is the essential object. Guessing a minimal annihilator from the exact ν sequence by the method of differential approximants [2, 12] returns an operator L of order four and degree fifteen.

This L is reducible in a trivial way: it admits the con-

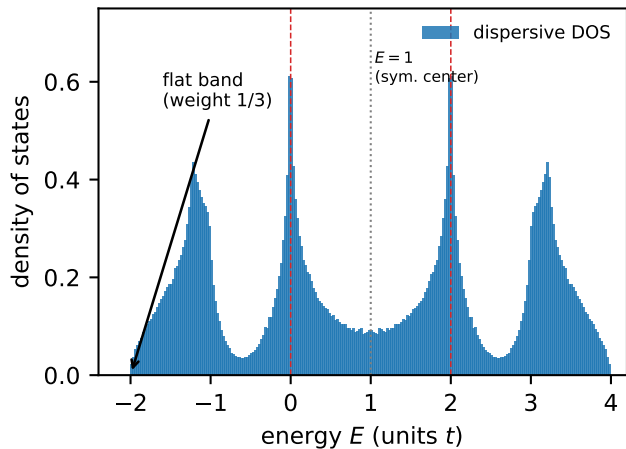


FIG. 1. Density of states of the hyperkagome tight-binding model ($t = 1$). Four flat bands at $E = -2$ carry weight $1/3$; the eight dispersive bands fill $[-2, 4]$ and are *exactly* symmetric about $E = 1$, with principal van Hove singularities at $E = 0, 2$ (dashed) and weaker interior features at $E = -1, 3$ and $E = 1 \pm \sqrt{5}$ (see text).

stant solution, so $L = M d/dt$ exactly, with M of order *three* and degree fifteen,

$$M = \sum_{i=0}^3 c_i(t) \left(\frac{d}{dt} \right)^i, \quad \deg c_i \leq 15, \quad (2)$$

the minimal operator annihilating Φ' . Its leading coefficient factorises as

$$c_3(t) \propto t^2 (t - \frac{1}{9})(t - \frac{1}{5})(t - \frac{1}{4})(t - 1)^3 p_7(t), \quad (3)$$

with p_7 an irreducible degree-seven polynomial. The finite singular points are images of the band structure under $t = (E - 1)^{-2}$: the band edge $E = 4$ (equivalently

$E = -2$) maps to $t = \frac{1}{9}$, the principal van Hove energies $E = 0, 2$ to $t = 1$, and the interior van Hove features $E = -1, 3$ and $E = 1 \pm \sqrt{5}$ to $t = \frac{1}{4}$ and $t = \frac{1}{5}$; the half-integer exponent in (4) confirms a square-root singularity at each. The remaining singular points ($t = 0$, $t = \infty$, and the degree-seven locus p_7) all appear in the complete Riemann scheme (4) below.

Certification. The operator M was reconstructed exactly over $\mathbb{Q}$ and shown to annihilate the series $\Phi'(t)$ *exactly* over $\mathbb{Q}$ through order t^{111} —112 independent relations against the 57 free coefficients of M , an overdetermination margin of 55 with zero residuals—a verification we confirmed modulo three independent large primes. It was furthermore re-derived from an entirely independent rebuild of the lattice (an independent quarter-depletion construction, independent closed-walk moment enumeration to m_{240} , and an independent guess): the moment sequence and all four coefficient polynomials of M are bit-identical. As a further independent check, the integer moments were reproduced by direct diagonalisation of the 12×12 Bloch Hamiltonian on a Brillouin-zone grid—a method disjoint from the closed-walk enumeration. This is the standard “conjectured-then-verified” certification of the lattice-statistics literature [2]. An unconditional derivation by creative telescoping [19, 20] from the rational integrand of the resolvent—which would upgrade this verification to a formal proof—is a natural next step, well suited to a dedicated holonomic-functions implementation.

IV. RIEMANN SCHEME AND DIFFERENTIAL-GALOIS STRUCTURE

The analytic nature of the LGF is decided by the local structure of M . Computing the Frobenius indicial exponents at each regular singular point gives the Riemann scheme

	$t = 0$	$t = \frac{1}{9}$	$t = \frac{1}{5}$	$t = \frac{1}{4}$	$t = 1$	p_7	$t = \infty$
exp.	$\{-1, -1, 0\}$	$\{-\frac{1}{2}, 0, 1\}$	$\{-\frac{1}{2}, 0, 1\}$	$\{-\frac{1}{2}, 0, 1\}$	$\{-1, -\frac{1}{2}, \frac{1}{2}\}$	$\{0, 1, 3\}$	$\{\frac{3}{2}, 2, 3\}$

(4)

Here p_7 denotes the irreducible degree-seven locus, whose seven Galois-conjugate roots all carry the same integer exponents $\{0, 1, 3\}$, computed exactly by reduction modulo p_7 . These are *apparent* singularities: integer exponents alone do not establish this, but all three local solutions there are log-free, which we verify from the Frobenius obstruction at each root of p_7 over several primes. We use the standard Riemann-scheme convention at infinity, in which an exponent ρ at $t = \infty$ corresponds to a local solution $y \sim t^{-\rho}$; the three solutions there thus behave as $t^{-3/2}$, t^{-2} and t^{-3} . As a check on (4), the operator has $s = 13$ singular points—the five rational

points above, the seven roots of p_7 , and $t = \infty$; the relation counts the apparent singularities on p_7 along with the rest—and the exponents sum to

$$-2 + 3 \cdot \frac{1}{2} - 1 + 7 \cdot 4 + \frac{13}{2} = 33 = \frac{n(n-1)}{2} (s-2), \quad n = 3, \quad (5)$$

so the Fuchs relation is satisfied exactly.

The repeated exponent -1 at $t = 0$ is the source of the logarithmic solution used below. The half-integer exponent $-\frac{1}{2}$ at the band edge $t = \frac{1}{9}$ is the square-root van Hove singularity of a three-dimensional band edge, $\rho(E) \sim \sqrt{E_{\text{edge}} - E}$.

We first record two local-to-global facts about M —it is

irreducible, and it is not a *literal* symmetric square—and then determine its differential Galois group, which is the object that actually decides the closed-form question.

M is irreducible over $\overline{\mathbb{Q}}(t)$. M is Fuchsian—regular singular at every point, including $t = \infty$ (exponents $\{\frac{3}{2}, 2, 3\}$)—so any first-order right factor $d/dt - u$ has $u = \sum_s e_s/(t - s)$ with e_s a local exponent at each singular point s and no polynomial part. At the rational singular points the e_s are the exponents of (4); at the irreducible degree-seven locus p_7 the common (Galois-invariant) residue ranges over its exponents $\{0, 1, 3\}$. This candidate set is finite, and each candidate is tested exactly over $\mathbb{Q}$ by the divisibility criterion $c_3(u'' + 3uu' + u^3) + c_2(u' + u^2) + c_1u + c_0 = 0$. No first-order right factor of M exists over $\mathbb{Q}(t)$, and none of $\text{adjoint}(M)$ —which for a third-order operator excludes an order-two right factor of M as well—so M has no right factor of order one or two over $\mathbb{Q}(t)$. We now descend to $\overline{\mathbb{Q}}(t)$ in two steps. *First*, M has no order-one right factor there: a hyper-exponential solution $d/dt - u$ with $u \in \overline{\mathbb{Q}}(t)$ has a finite Galois orbit $\{d/dt - u^\sigma\}$ of right factors of M , whose least common left multiple is a right factor defined over $\mathbb{Q}(t)$ of order $k \in \{1, 2, 3\}$; $k = 1$ gives $u \in \mathbb{Q}(t)$ and $k = 2$ an order-two right factor over $\mathbb{Q}$, both excluded above, while $k = 3$ would make M the LCLM of order-one operators, hence completely reducible, forcing *semisimple* local monodromy and no logarithm—contradicting the genuine logarithm at $t = 0$ (repeated indicial exponent -1 , exactly one log-free local solution of three, verified directly). *Second*, M has no order-two right factor over $\overline{\mathbb{Q}}(t)$: such a factor is equivalent to an order-one right factor of $\text{adjoint}(M)$ over $\overline{\mathbb{Q}}(t)$, excluded by the identical descent applied to $\text{adjoint}(M)$ (no order-one factor over $\mathbb{Q}$ by the adjoint search, no order-two by the M search, and $\text{adjoint}(M)$ is not completely reducible, since complete reducibility is self-dual and M is not). Hence M is irreducible over $\overline{\mathbb{Q}}(t)$. (This exact enumeration replaces the floating-point candidate search of earlier drafts; the computation is provided in the Supplemental Material and is independently reproducible with a certified factorizer such as Maple's **DFactor** or Magma.)

M is not a literal symmetric square. If M were literally the symmetric square of a second-order operator with local exponents $\{a, b\}$, its three exponents would be $\{2a, a + b, 2b\}$, an arithmetic progression whose middle term is the mean of the outer two. No triple in (4) is of this form: at $t = \frac{1}{9}$ the exponents $\{-\frac{1}{2}, 0, 1\}$ have mean of outer terms $\frac{1}{4} \neq 0$, and at $t = 1$ the exponents $\{-1, -\frac{1}{2}, \frac{1}{2}\}$ have mean of outer terms $-\frac{1}{4} \neq -\frac{1}{2}$; a single such point already suffices. As a positive control, the same analysis applied to the simple-cubic LGF returns the symmetric-square-compatible exponents $\{0, \frac{1}{2}, 1\} = \{2a, a + b, 2b\}$ with $\{a, b\} = \{0, \frac{1}{2}\}$, correctly reflecting its closed form in K . *The scope of this exclusion is limited, and it is essential not to overstate it.* A gauge transformation shifts the three exponents at each point by a common constant, and an algebraic pullback multiplies them by the local ramification index; both preserve

arithmetic progressions, so the exponent test also rules out every *function-multiplier* or *pullback* equivalent of a symmetric square. It does *not*, however, rule out an equivalence realised by a *differential intertwiner*—a homomorphism of differential modules of order two, which shifts the individual local exponents by *different* integers and so destroys the arithmetic progressions while preserving the Galois-theoretic content. This is exactly the loophole the hyperkagome operator exploits: as we show in Sec. V, M is equivalent to a symmetric square through such an order-two intertwiner, even though it is not a literal symmetric square (nor a multiplier/pullback of one). Correcting an overstatement in an earlier version of this work, the not-a-literal-symmetric-square property therefore does *not* exclude an elliptic closed form; the closed-form question is settled instead by the Galois group, which we now compute.

No algebraic or elementary closed form. On a separate axis, M has no *Liouvillian* solutions, hence no closed form in algebraic or elementary functions. By the Singer–Ulmer classification, an irreducible third-order operator has Liouvillian solutions only if its differential Galois group is finite or imprimitive. A finite group makes every solution algebraic, hence log-free. An imprimitive group permutes three lines, so in the corresponding basis every element—in particular every local monodromy, since for a Fuchsian operator the Galois group is the Zariski closure of the monodromy group—is a monomial matrix. Such a matrix is necessarily semisimple: if its permutation part has order k , its k -th power is diagonal and invertible, and an invertible matrix with a diagonalisable power is itself diagonalisable in characteristic zero. In either case there can be *no logarithmic solution*. But M has a genuine logarithmic solution: at $t = 0$ the indicial exponent -1 is repeated and only one log-free (Laurent) local solution exists (of three), so the local monodromy is non-semisimple. This excludes both the finite and the imprimitive cases, so the Galois group acts irreducibly and non-solvably and M is non-Liouvillian. The differential-Galois computation of the next section gives the same conclusion more sharply: $G^\circ = \text{SO}(3, \mathbb{C})$ is simple, hence non-solvable, and an irreducible operator has Liouvillian solutions if and only if G° is solvable. Being non-Liouvillian does *not* preclude an elliptic form—the complete elliptic integral K is itself non-Liouvillian, and the simple-cubic LGF operator is irreducible and non-Liouvillian yet closed-form in K . Indeed, as the next section shows, the hyperkagome operator *is* orthogonally equivalent to a symmetric square, and Sec. VI proves the resulting closed form outright—it is modular, through eta quotients and Hauptmoduln, which are themselves non-Liouvillian; the non-Liouvillian result excludes only the algebraic and elementary cases and is fully consistent with modularity.

V. AN INVARIANT QUADRATIC FORM: THE ORTHOGONAL GALOIS GROUP

The symmetric square $\text{Sym}^2(M)$ —the order-six operator annihilating the pairwise products $y_i y_j$ of solutions

$$(c_{11}, c_{12}, c_{13}, c_{22}, c_{23}, c_{33}) = \left(-\frac{17}{84}, -\frac{460}{441}, \frac{44927}{3087}, -\frac{1925200}{157437}, \frac{83066560}{1102059}, -\frac{712021969}{7714413} \right), \quad \det C = \frac{3437476900}{7714413} \neq 0 \text{ (rank 3)}. \quad (7)$$

Equation (6) is verified as an exact identity: the six constants are fixed by six series coefficients and then satisfy 194 further exact relations at each of two independent ordinary base points ($t_0 = \frac{1}{2}$ and $t_0 = -\frac{1}{3}$). The agreement is a proof, not merely a margin, by a Fuchs-relation budget: were $F = R - \sum c_{ij} y_i y_j$ not identically zero, the span of the six products and R would be seven-dimensional, with a Fuchsian minimal operator of order seven whose local exponents are bounded below by the certified data—the pairwise sums of the exponents of (4) at each singular point, the pole and zero orders of R , and *distinct* non-negative integers at every apparent point. The Fuchs relation then caps the vanishing order of any nonzero element of that span at an M -ordinary point at 109, while F is verified to vanish to order 200; hence $F \equiv 0$ identically. Independently and at the operator level, we exhibit an order-two differential intertwiner $T \in \text{Hom}(\text{adjoint}(M), M)$ and verify the exact identity $\text{rightrem}(MT, \text{adjoint}(M)) = 0$ over $\mathbb{Q}(t)$, so M is homomorphic—in fact isomorphic, since both are irreducible of the same rank—to its adjoint. Four consequences follow.

(1) *The monodromy lies in an orthogonal group.* Because R is single-valued (rational) and $R = \sum c_{ij} y_i y_j$ with the six products linearly independent (the leading 6×6 coefficient system is nonsingular), every monodromy matrix γ obeys $\gamma^\top C \gamma = C$: the solution space carries a monodromy-invariant symmetric bilinear form. M is Fuchsian, so by the density theorem the differential Galois group G is the Zariski closure of the monodromy group, whence $G \subseteq \text{O}(C) \cong \text{O}(3, \mathbb{C})$.

(2) *The form is nondegenerate.* By irreducibility, the kernel of C —an invariant subspace—is 0, so C is nondegenerate; by Schur's lemma the invariant form is unique up to scale, and in odd dimension it is symmetric rather

than symplectic. We confirm $\det C \neq 0$ directly, at both base points.

$$R(t) = -\frac{1}{272} \frac{(15t^2 + 17t - 8)^2}{t^2(t-1)^2(4t-1)(5t-1)(9t-1)}. \quad (6)$$

Expanding $R = \sum_{i \leq j} c_{ij} y_i y_j$ in a fundamental system y_1, y_2, y_3 of M determines a symmetric 3×3 Gram matrix $C = (c_{ij})$. In the basis fixed by the initial conditions $y_i^{(j)}(t_0) = \delta_{ij}$ at the ordinary point $t_0 = \frac{1}{2}$,

than symplectic. We confirm $\det C \neq 0$ directly, at both base points.

(3) *The identity component is $\text{SO}(3, \mathbb{C})$.* The genuine logarithm at $t = 0$ makes a monodromy element non-semisimple, so G is infinite. Suppose, for contradiction, that G° were solvable. As a normal subgroup of G , G° acts semisimply on the solution space V by Clifford's theorem (since V is G -irreducible); by the Lie–Kolchin theorem the irreducible representations of a connected solvable group are one-dimensional, so G° acts diagonally in some basis and G permutes the resulting isotypic components. If the three characters are distinct, G is monomial (imprimitive)—but every monomial matrix is semisimple (if its permutation part has order k , its k -th power is diagonal and invertible, and an invertible matrix with a diagonalisable power is diagonalisable in characteristic zero), contradicting the logarithm. A $(2+1)$ pattern of characters cannot be permuted nontrivially, which would make V reducible—again a contradiction. A single character makes G° scalar, hence, inside $\text{O}(3, \mathbb{C})$, finite, forcing G finite and every element semisimple—once more contradicting the logarithm. Hence G° is non-solvable, and the only non-solvable connected subgroup of $\text{O}(3, \mathbb{C})$ is $\text{SO}(3, \mathbb{C})$; therefore $G^\circ = \text{SO}(3, \mathbb{C})$.

(4) *The determinant character: $G = \text{O}(3, \mathbb{C})$ in full.* The Wronskian Wr of M satisfies $\text{Wr}'/\text{Wr} = -c_2/c_3$, whose residue at each singular point equals the local exponent sum there minus 3. This is half-integral exactly at $t = \frac{1}{9}, \frac{1}{5}, \frac{1}{4}$ and ∞ (for instance at $\frac{1}{9}$, $(-\frac{1}{2} + 0 + 1) - 3 = -\frac{5}{2}$) and integral at $t = 0, 1$ and on p_7 ; hence $\det(\text{monodromy}) = -1$ at those four points and $+1$ elsewhere. Thus G is *not* contained in $\text{SO}(3, \mathbb{C})$, i.e. $G = \text{O}(3, \mathbb{C})$ in full, and the determinant character is the quadratic character of the genus-one curve

$$v^2 = (1 - 4t)(1 - 5t)(1 - 9t), \quad (8)$$

ramified at exactly $\{\frac{1}{9}, \frac{1}{5}, \frac{1}{4}, \infty\}$ —an even number of points, as a double cover requires, and the same four points at which the pole orders of R in (6) are odd.

Since $\text{SO}(3, \mathbb{C}) \cong \text{PSL}(2, \mathbb{C})$ through the symmetric square of the standard two-dimensional representation, $G^\circ = \text{SO}(3, \mathbb{C})$ means that M is *projectively equivalent*

¹ The homomorphism of M to its adjoint, the rational solution R of $\text{Sym}^2(M)$, and the explicit order-two intertwiner realising it are due to J.-M. Maillard (private communication, July 2026); we reproduce and certify them independently in exact arithmetic in the accompanying repository.

to $\text{Sym}^2(V_2)$ for a second-order operator V_2 (defined over $\mathbb{Q}(t, v)$ with v as in (8), or over $\mathbb{Q}(t)$ after the corresponding quadratic twist). The equivalence is carried by the order-two intertwiner above; it shifts individual local exponents by *different* integers and so destroys the arithmetic progressions, which is exactly why M is not a literal symmetric square (nor a multiplier or pullback of one) yet is orthogonally, projectively, one. The identical structure is known in the differential algebra of lattice models: Assis, van Hoeij and Maillard [13] meet an order-three operator N_3 homomorphic to a symmetric square $\text{Sym}^2(V_2)$, recover V_2 by a conic construction, and solve it as a ${}_2F_1$ with an algebraic (modular) pullback. For the hyperkagome that conic construction has now been carried out in the accompanying analysis: the transported invariant form admits an explicit rational conic point, yielding an *explicit* second-order operator V_2 over $\mathbb{Q}(t)$ (the quadratic twist enters only through the pullback), verified in exact arithmetic through the bridge identity $f_0^2 = P(y_0)$ with $V_2(f_0) = 0$ to more than a hundred series orders, where P is an explicit order-two intertwiner and $y_0 = \Phi'/2$ the analytic solution. The pullback question is settled, in its sharpest form, in Sec. VI: V_2 is the uniformizing operator of the modular curve $X(\Gamma_0(30)^+)$, its mirror map at the MUM point is the level-30 eta-quotient Hauptmodul (11), and the sought closed form is modular—an Assis–van Hoeij–Maillard-style ${}_2F_1$ with *algebraic* pullback exists in principle because j is algebraic over $\mathbb{Q}(t)$, but the modular parametrization itself is the closed form in its natural coordinates.

Maximal unipotent monodromy at $t = 0$ ($n = 2$). The three exponents at $t = 0$, $\{-1, -1, 0\}$ (indicial polynomial $-64\rho(\rho + 1)^2$), are all integers, so the local monodromy is unipotent. Exactly one of the three local solutions is log-free (single-valued Laurent); since the number of log-free solutions equals the number of Jordan blocks of a unipotent, the monodromy is a *single* 3×3 block: the point is *maximally unipotent* (MUM), with maximal log power $n = 2$. It is MUM in the mon-

odromy sense with exponents $\{-1, -1, 0\}$ rather than $\{0, 0, 0\}$: an integer recombination of the Frobenius solutions brings the basis to the canonical MUM normal form $\{y_0, y_0 \log t + f_1, \frac{1}{2}y_0 \log^2 t + f_1 \log t + f_2\}$, with the *same* series f_1 multiplying the logarithm in the third solution as appears in the second,² where $y_0 = \Phi'/2$ is the holomorphic solution and f_1, f_2 carry $1/t$ Laurent heads. This settles a point that was initially unclear: $n = 2$, not 1. The even value $n = 2$ is consistent with the orthogonal (rather than symplectic) case of the parity correspondence of Hassani, Maillard and Zenine [14], in which the parity of the maximal log power tracks the orthogonal/symplectic type of the differential Galois group. In the energy variable $t = (E - 1)^{-2}$ the point $t = 0$ is $E = \infty$ (equivalently $z = \infty$), so this is the local expansion at infinity, and the $1/t$ Laurent heads of the logarithmic partners are a feature of that point rather than a pathology.

VI. THE MODULAR PARAMETRIZATION: V_2 UNIFORMIZES $X(\Gamma_0(30)^+)$

We now prove the statement toward which Secs. V and VIII point: the second-order operator V_2 is the *uniformizing differential equation* of a classical modular curve, and the variable t is a Hauptmodul. Throughout, τ lies in the upper half-plane, $q = e^{2\pi i \tau}$, $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$, and $\Gamma_0(30)^+$ denotes the extension of $\bar{\Gamma}_0(30)$ by its seven Atkin–Lehner involutions w_e , $e \in \{2, 3, 5, 6, 10, 15, 30\}$ [21, 23]; we henceforth write v for the twist variable of Eq. (8), reserving u for the modular function below. The projective normal form of the explicit V_2 of Sec. V—the operator $w'' + Q_V(t)w = 0$ obtained from $V_2 = \partial_t^2 + a_1 \partial_t + a_0$ via $Q_V = a_0 - \frac{1}{4}a_1^2 - \frac{1}{2}a_1'$, invariant under every function-multiplier gauge and hence a projective invariant of M itself—is

$$Q_V(t) = \frac{N(t)}{4t^2(t-1)^2(4t-1)^2(5t-1)^2(9t-1)^2}, \quad N(t) = 24300t^8 - 58860t^7 + 73437t^6 - 44294t^5 + 15111t^4 - 3160t^3 + 407t^2 - 30t \quad (9)$$

The double poles of Q_V sit *exactly* at the six points $t = 0, \frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1, \infty$: the seven apparent singularities on p_7 , and every trace of the order-two intertwiner, cancel identically in the normal form. The Laurent head of Q_V at each of $t = \frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1, \infty$ is $\frac{3}{16}$ (local exponent differ-

ence $\frac{1}{2}$), and at $t = 0$ it is $\frac{1}{4}$ (exponent difference 0: the MUM point). Define the level-30 eta quotient

$$u(\tau) = \prod_{d|30} \eta(d\tau)^{r_d} = \left[\frac{\eta(\tau) \eta(6\tau) \eta(10\tau) \eta(15\tau)}{\eta(2\tau) \eta(3\tau) \eta(5\tau) \eta(30\tau)} \right]^3, \quad (10)$$

i.e. $(r_d) = (3, -3, -3, -3, 3, 3, 3, -3)$ on $d = (1, 2, 3, 5, 6, 10, 15, 30)$, and set

$$t = \frac{u}{u^2 + 7u + 1}, \quad \text{equivalently} \quad \frac{1}{t} = u + 7 + \frac{1}{u}, \quad (11)$$

² The recombination ($yy_2 = y_2 - \frac{13}{30}y_1$ in the natural rescaling) was observed by J.-M. Maillard (private communication, July 2026) and is verified exactly, term by term and through the identity $M(y_0 \log t + f_1) = 0$ order by order, in `verify_mum_normalform.py` in the repository.

so that $t = q - 4q^2 + 12q^3 - 34q^4 + 90q^5 - \dots$.

Theorem. $1/t$ is a Hauptmodul of $\Gamma_0(30)^+$ (so t is a Hauptmodul vanishing at the unique cusp), and the Schwarzian identity

$$\{\tau, t\} = 2Q_V(t), \quad \{\tau, t\} \equiv \frac{\tau'''}{\tau'} - \frac{3}{2} \left(\frac{\tau''}{\tau'} \right)^2, \quad ' = \frac{d}{dt}, \quad (12)$$

holds exactly [28]. Equivalently: the ratio of two solutions of $w'' + Q_V w = 0$ inverts the Hauptmodul, V_2 is the uniformizing operator of the orbifold $X(\Gamma_0(30)^+)$, and the mirror map of V_2 at its MUM point $t = 0$ is precisely (11), with the modular q as nome. (The identification was found by matching the mirror map against (10)–(11) to all 96 computed orders; the proof below shows the match is forced.)

Proof, modular part. The exponent vector (r_d) satisfies Ligozat's criterion [22]: $\sum_d d r_d = -24 \equiv 0$ and $\sum_d (30/d) r_d = 24 \equiv 0 \pmod{24}$, the weight $\frac{1}{2} \sum_d r_d$ vanishes, and $\prod_d d^{r_d} = 1$ is a rational square; hence u is a modular function on $\Gamma_0(30)$. Ligozat's cusp-order formula gives its divisor on $X_0(30)$ (cusps labelled by their denominators $d \mid 30$): simple zeros at the cusps $\{1, 6, 10, 15\}$ and simple poles at $\{2, 3, 5, 30\}$, so u has degree four. For the Fricke involution, the exact antisymmetry $r_{30/d} = -r_d$ forces $w_{30}(u) = 1/u$ structurally (the eta multipliers cancel because $\sum_d r_d = 0$ and $\prod_d (30/d)^{r_d} = 1$); comparing divisors and leading Fourier coefficients—two modular functions with equal divisors and equal leading coefficient are equal—shows likewise that w_2, w_3, w_5 each send $u \mapsto 1/u$, whence the complementary products w_6, w_{10}, w_{15} fix u . Since $\{1, w_6, w_{10}, w_{15}\}$ is an index-two subgroup of the Atkin-Lehner group $\mathcal{W} \cong (\mathbb{Z}/2)^3$, u descends to a function of degree $4/4 = 1$ on $X(\Gamma_0(30)+6, 10, 15)$: it is a Hauptmodul there, and that curve has genus zero. Consequently $1/t = u + 1/u + 7$ is invariant under all of $\Gamma_0(30)^+$ and descends to degree one on the quotient: it is a Hauptmodul of $\Gamma_0(30)^+$, and $X(\Gamma_0(30)^+)$ has genus zero. It has a single cusp: w_e maps the cusp of denominator d to that of denominator $de/\gcd(d, e)^2$, a simply transitive action of $\mathcal{W}$ on the eight cusps of $X_0(30)$. The normalization (11) puts the cusp at $t = 0$, with $t = q + O(q^2)$ (cusp width one).

Proof, uniformization part. First, the signature of $\Gamma_0(30)^+$ is forced. $\Gamma_0(30)$ is torsion-free: for $30 = 2 \cdot 3 \cdot 5$ the elliptic-point counts are $\nu_2 = \prod_{p \mid 30} (1 + (\frac{-1}{p})) = 0$ (the factor at $p = 3$ vanishes) and $\nu_3 = \prod_{p \mid 30} (1 + (\frac{-3}{p})) = 0$ (the factor at $p = 2$ vanishes) [24]. Any elliptic $\gamma \in \Gamma_0(30)^+$ has $\gamma^2 \in \Gamma_0(30)$, and γ^2 is elliptic or the identity; torsion-freeness forces $\gamma^2 = 1$, so every elliptic point of $\Gamma_0(30)^+$ has order two. The orbifold Euler characteristic is $\chi = \frac{1}{8} \chi(\Gamma_0(30)) = -\frac{1}{8} \cdot \frac{72}{6} = -\frac{3}{2}$ (the index is $[\mathrm{PSL}_2(\mathbb{Z}) : \Gamma_0(30)] = 72$), i.e. hyperbolic covolume 3π —nine times that of $\mathrm{PSL}_2(\mathbb{Z})$, a covolume ratio and not a subgroup index, since $\Gamma_0(30)^+$ is not a subgroup of $\mathrm{PSL}_2(\mathbb{Z})$. With genus zero and one cusp, Gauss–Bonnet $2 - 1 - \sum_i (1 - e_i^{-1}) = -\frac{3}{2}$ gives exactly five elliptic points,

all of order two: the signature $(0; 2, 2, 2, 2, 2; 1 \text{ cusp})$ is derived, not assumed. Second, the identity (12) follows from a degree count. Because t is a Hauptmodul of a finite-covolume Fuchsian group, $\{\tau, t\}$ is a rational function of t , with double poles precisely at the images of the elliptic points and of the cusp, and with universal Laurent heads: $\frac{1}{2}(1 - e^{-2})$ at an order- e elliptic image and $\frac{1}{2}$ at a cusp image—so $\frac{1}{2}\{\tau, t\}$ carries heads $\frac{3}{16}$ and $\frac{1}{4}$, matching (9). Consider $D(t) = \{\tau, t\} - 2Q_V(t)$, a rational function. Its poles are confined to the five (a priori unlocated) elliptic images, the cusp image $t = 0$, the remaining finite poles $\frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1$ of Q_V , and $t = \infty$, each of order at most two; at $t = 0$ the heads $\frac{1}{2t^2}$ cancel exactly, leaving at most a simple pole. The pole divisor of D on $\mathbb{P}^1$ therefore has degree at most $2(5+4+1)+1 = 21$. On the other hand we verify, in exact rational arithmetic, that the q -expansion of $D(t(q))$ —computable term by term from (10)–(11)—vanishes through order q^{96} ; since $t(q) = q + O(q^2)$ is a formal immersion, D vanishes at $t = 0$ to order at least 96. A nonzero rational function has equal degrees of zeros and poles, and $96 > 21$; hence $D \equiv 0$. $\square$

The proof is exact and non-numerical throughout: the modular part is finite eta-quotient combinatorics, and the uniformization part is a finite exact series computation closed off by the degree bound. As a corollary the double poles of $\{\tau, t\} = 2Q_V$ are exactly $t = 0, \frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1, \infty$, so the five order-two points of $\Gamma_0(30)^+$ project precisely to $\{\frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1, \infty\}$ and the cusp to $t = 0$: the singular locus of V_2 (equivalently of M , up to the apparent p_7) is the orbifold locus of $X(\Gamma_0(30)^+)$, with no elliptic point hiding elsewhere. In u -coordinates the dictionary is explicit: $t = \frac{1}{9} \leftrightarrow u = 1$ and $t = \frac{1}{5} \leftrightarrow u = -1$ (the branch points of $u \mapsto u + 1/u$), while $t = \frac{1}{4}, 1, \infty$ correspond to the root pairs of $u^2 + 3u + 1$, $u^2 + 6u + 1$, $u^2 + 7u + 1$, and the cusp $t = 0$ to $u \in \{0, \infty\}$.

Three consequences deserve separate statement. (i) *The monodromy is arithmetic.* The projective monodromy group of V_2 —hence of M , by the projective equivalence of Sec. V—is the deck group of the uniformization: exactly the discrete arithmetic lattice $\Gamma_0(30)^+ \subset \mathrm{PSL}_2(\mathbb{R})$, of covolume 3π . (ii) *The twist is itself modular.* The determinant character of Sec. V is a quadratic character of $\Gamma_0(30)^+$ trivial on $\Gamma_0(30)$; its kernel is the index-two subgroup $\Gamma_0(30)+2, 3, 6$, and the associated double cover of the t -line, branched at the four $\det = -1$ points $\{\frac{1}{9}, \frac{1}{5}, \frac{1}{4}, \infty\}$, has genus one by Riemann–Hurwitz—it is precisely the twist curve $v^2 = (1 - 4t)(1 - 5t)(1 - 9t)$ of Eq. (8), now identified as $X(\Gamma_0(30)+2, 3, 6)$. In particular the analytic period $y_0 = \Phi'/2$, pulled back by (11), is a weight-two object carrying this quadratic twist character, in the manner familiar from the modular interpretation of the Apéry numbers [25]—though, as Sec. VII shows, a depth-one quasimodular one rather than a modular form; its explicit closed form is derived there. (iii) *The Varma–Monien question is resolved.* Varma and Monien judged it “highly suggestive that a closed form expression can be obtained” for their threefold integral [10]; the theo-

rem proves this in the strongest structural sense, with the closed form modular rather than elliptic: the hyperkagome LGF lives at level $30 = 2 \cdot 3 \cdot 5$. To our knowledge every lattice Green's function (and sunrise-integral period) previously realised modularly occurs at a level with at most two distinct primes—levels 2, 3, 4, 6 dominate the classical cubic-lattice and Bessel-moment literature [26, 27]—so the hyperkagome appears to be the first lattice Green's function realised at a level with three distinct prime factors. An Assis–van Hoeij–Maillard-style ${}_2F_1$ representation [13] follows in principle, since j is algebraic over $\mathbb{Q}(t)$ of bounded degree (an algebraic, not rational, pullback); but (10)–(11) is already the closed form in its most natural coordinates.

VII. THE WEIGHT-TWO PERIOD IN CLOSED FORM

The weight-two period $y_0 = \Phi'/2$ itself admits an explicit closed form in the modular data of Sec. VI. Write $W = q dt/dq$ for the nome derivative of the Hauptmodul (11) (so that W is a constant multiple of w_1^2 , with $w_1 = (dt/d\tau)^{1/2}$ a solution of the projective normal form $w'' + Q_V w = 0$ of (9)), put $W' = dW/dt$, and let v be the twist unit of Eq. (8) in the branch $v(0) = +1$. Then

$$y_0 = \frac{\rho_0(t) W + \rho_1(t) W'}{v}, \quad (13)$$

$$\rho_1(t) = \frac{15t^2 + 17t - 8}{30t(t-1)}, \quad \rho_0(t) = -\frac{4050t^6 + 2445t^5 - 11436t^4 + 8000t^3 - 2130t^2 + 231t - 8}{30t^2(t-1)^2(4t-1)(5t-1)(9t-1)}. \quad (14)$$

Equivalently, cleared of all denominators,

$$B(t) y_0 v W = A_0(t) W^2 + A_1(t) q \frac{dW}{dq}, \quad (15)$$

with $B = -30t^2(t-1)^2(4t-1)(5t-1)(9t-1)$, $A_0 = 4050t^6 + 2445t^5 - 11436t^4 + 8000t^3 - 2130t^2 + 231t - 8$,

and $A_1 = -t(t-1)(4t-1)(5t-1)(9t-1)(15t^2 + 17t - 8)$, using $q dW/dq = WW'$. Every ingredient is an explicit level-30 modular or quasimodular object: with u the eta quotient (10), $E_2(\tau) = 1 - 24 \sum_{n \geq 1} \sigma_1(n) q^n$, and the Eisenstein form of the η -logarithmic derivative

$$\ell = q \frac{d \log u}{dq} = \frac{1}{8} [E_2(\tau) + 6E_2(6\tau) + 10E_2(10\tau) + 15E_2(15\tau) - 2E_2(2\tau) - 3E_2(3\tau) - 5E_2(5\tau) - 30E_2(30\tau)], \quad (16)$$

one has $W = (1 - u^2) u \ell / (u^2 + 7u + 1)^2$ and $v = (u^2 - 1) \sqrt{u^2 + 3u + 1} / (u^2 + 7u + 1)^{3/2}$.

Proof sketch. The identity is proven, not fitted, in five steps. [I] By the Schwarzian identity (12), $w_1 = (dt/d\tau)^{1/2}$ solves $w'' + Q_V w = 0$, so W solves its symmetric square. [II] Elementarily, $y'' = -Q_V y$ forces $z = y^2$ to satisfy $\tilde{N}(z) := z''' + 4Q_V z' + 2Q_V' z = 0$; hence $\tilde{N}(W) = 0$. [III] Conjugating the hyperkagome operator by the twist unit, $M_v = v \circ M \circ v^{-1}$ has coefficients in $\mathbb{Q}(t)$ (the logarithmic derivative v'/v is rational) and annihilates $y_0 v$. [IV] The first-order operator $T = \rho_0 + \rho_1 d/dt$ satisfies the exact operator identity $M_v T = X \tilde{N}$ in the ring $\mathbb{Q}(t)\langle d/dt \rangle$ of linear differential operators over $\mathbb{Q}(t)$, with right remainder identically zero and X of order one whose top-order coefficient carries the apparent-singularity polynomial p_7 ; therefore T maps solutions of $\tilde{N}$ to solutions of M_v , and $T(W)$ solves M_v . [V] $T(W)$ is single-valued near $t = 0$ (W and W' are single-valued functions of t there, since $t(q) = q + O(q^2)$ is invertible), and exactly one of the three local solutions of M at the maximal-unipotent point $t = 0$ is log-free (Sec. V); since $v(0) = 1 \neq 0$, conjugation by v preserves

this local structure, so the single-valued solution line of M_v at $t = 0$ is one-dimensional and spanned by $y_0 v$, and matching a single series coefficient gives $y_0 v = T(W)$. As an independent check, (13) is verified as an exact q -series through order q^{171} , assembled entirely from the raw η and E_2 definitions above.

Three structural remarks follow. (i) y_0 is *quasimodular, not modular*. It is a weight-two *depth-one* quasimodular form on $\Gamma_0(30)^+$ carrying the orthogonal determinant character of Sec. V; equivalently it lives on the genus-one twist double cover $X(\Gamma_0(30)+2, 3, 6)$ of Eq. (8). (ii) *The derivative term is essential*. In the minimal-degree representation (15)—unique, as the fitting nullspace is one-dimensional—the coefficient A_1 is not identically zero: the term $\rho_1 W'$ is exactly the quasimodular depth introduced by inverting the order-two intertwiner of Sec. V. A depth-zero *rational* form $y_0 v = R(t) W$ is excluded outright by local exponents: at the elliptic point $t = \frac{1}{9}$ both W and the twist unit v vanish to order $\frac{1}{2}$, while y_0 carries the exponent $-\frac{1}{2}$ of M 's Riemann scheme there (the square-root van Hove singularity, $[t^m] y_0 \sim 9^m m^{-1/2}$), so $y_0 v$ is regular at $t = \frac{1}{9}$

whereas $R(t)W$ has half-integer order for every rational R ; the exact scan of (15) through degree 40 agrees. In fact the exclusion is categorical: no algebraic factor, at any weight and any level, can absorb the derivative term. This rests on one group-theoretic fact.

Lemma. The differential Galois group $\mathcal{G}$ of $w'' + Q_V w = 0$ over $\mathbb{C}(t)$ is the full $\mathrm{SL}(2, \mathbb{C})$.

Proof. The Wronskian is constant, so $\mathcal{G} \subseteq \mathrm{SL}(2, \mathbb{C})$.

$$\frac{\tilde{N}(z)}{z} = r'' + 3rr' + r^3 + 4Q_V r + 2Q_V' = \frac{-30p_7(t)}{t^2(t-1)(4t-1)(5t-1)(9t-1)(15t^2+17t-8)^3} \quad (17)$$

—the apparent-singularity polynomial once more, and manifestly nonzero—shows $z \notin \mathrm{Sol}(\tilde{N})$. Hence T is a bijection $\mathrm{Sol}(\tilde{N}) \rightarrow \mathrm{Sol}(M_v)$; being defined over $\mathbb{Q}(t)$, it is equivariant for the differential Galois action, computed in a Picard–Vessiot field containing both solution spaces (whose Galois group restricts *onto* the Galois group of each operator [16]), so it identifies the Galois representation on $\mathrm{Sol}(M_v)$ with the symmetric square of the standard representation of $\mathcal{G}$. Now $\mathrm{Gal}(M_v)^\circ = \mathrm{Gal}(M)^\circ = \mathrm{SO}(3, \mathbb{C})$ (Sec. V): a finite extension of the base field replaces the Galois group by a closed subgroup of finite index, which shares its identity component [16]—applied once to M and once to M_v across $\mathbb{C}(t) \subset \mathbb{C}(t, v)$ —and over $\mathbb{C}(t, v)$ multiplication by the base-field element v identifies $\mathrm{Sol}(M_v) = v \mathrm{Sol}(M)$ equivariantly, with identical matrices. If $\mathcal{G}$ were a proper closed subgroup of $\mathrm{SL}(2, \mathbb{C})$, its identity component would be solvable (it lies in a Borel subgroup or in the normalizer of a torus, or is finite [16, 17]), and then so would the identity component $\mathrm{Sym}^2(\mathcal{G})^\circ \cong \mathrm{Gal}(M_v)^\circ$ of its symmetric square—contradicting the non-solvability of the simple group $\mathrm{SO}(3, \mathbb{C})$. Hence $\mathcal{G} = \mathrm{SL}(2, \mathbb{C})$. (Independently, the theorem of Sec. VI identifies the projective monodromy of $w'' + Q_V w = 0$ with the lattice $\Gamma_0(30)^+$, which is Zariski-dense in $\mathrm{PSL}(2, \mathbb{C})$, and the Galois group of a Fuchsian operator is the Zariski closure of its monodromy; either route suffices.)

Theorem (no modular-times-algebraic form at any weight). Let $F \neq 0$ be a meromorphic modular form of any weight $k \in \frac{1}{2}\mathbb{Z}$, with any finite-order multiplier system, on any subgroup $\Gamma' \subset \mathrm{PSL}(2, \mathbb{R})$ commensurable with $\Gamma_0(30)^+$, and let $a \neq 0$ be any function algebraic over $\mathbb{C}(t)$. Then $y_0 \neq Fa$ (as functions of τ under the parametrization (11)). In particular y_0 is not a pure eta quotient, of any level, nor a (weight-two modular form) $\times$ (algebraic function): every $\Gamma_0(N)$ and its Atkin–Lehner extensions are commensurable with $\Gamma_0(30)^+$, and the eta multiplier has finite order (dividing 24) [22, 24].

Proof. Suppose $y_0 = Fa$; multiplying by the algebraic twist unit v gives $y_0 v = F\tilde{a}$ with $\tilde{a} = av$ still algebraic and nonzero. Work in the Picard–Vessiot field K of $w'' + Q_V w = 0$ over $\mathbb{C}(t)$: by step [I], W is a constant

The solution space of $\tilde{N}$ is spanned by the products $w_i w_j$ of solutions of $w'' + Q_V w = 0$ [16], and by the operator identity of step [IV] the order-one operator $T = \rho_0 + \rho_1 d/dt$ maps $\mathrm{Sol}(\tilde{N})$ into $\mathrm{Sol}(M_v)$. Any kernel element of T is a constant multiple of $z = \exp(-\int \rho_0/\rho_1)$, and the exact rational computation (with $r = z'/z = -\rho_0/\rho_1$)

multiple of w_1^2 , so $\mathcal{L} = W'/W = 2w_1'/w_1 \in K$ and the closed form (13) reads $y_0 v = W(\rho_0 + \rho_1 \mathcal{L})$. Since t is $\Gamma_0(30)^+$ -invariant, $W \propto dt/d\tau$ is a weight-two meromorphic form with *trivial* multiplier on all of $\Gamma_0(30)^+$; both F and W have meromorphic q -expansions at every cusp of the commensurability class. Pick a positive integer n such that $nk \in 2\mathbb{Z}$ —making $(c\tau + d)^{nk}$ an honest integer-weight automorphy factor, free of the half-integer branch ambiguity, so that the n -th power of the multiplier system is a genuine finite-order character—and, enlarging n by the order of that character, such that it is trivial. Then $\Psi = F^n/W^{nk/2}$ is a weight-zero modular function with trivial multiplier on the finite-index subgroup $\Gamma' \cap \Gamma_0(30)^+$, meromorphic at the cusps, hence a rational function on the compact curve $X(\Gamma' \cap \Gamma_0(30)^+)$ —a finite cover of $X(\Gamma_0(30)^+)$ —and therefore algebraic over $\mathbb{C}(X(\Gamma_0(30)^+)) = \mathbb{C}(t)$. Raising $y_0 v = F\tilde{a}$ to the n -th power, inserting $F^n = \Psi W^{nk/2}$, and dividing by W^n gives

$$(\rho_0 + \rho_1 \mathcal{L})^n = \mathcal{A} W^{n(k-2)/2}, \quad \mathcal{A} = \Psi \tilde{a}^n \neq 0. \quad (18)$$

Every member of (18) lies in K , so $\mathcal{A} \in K$ is algebraic over $\mathbb{C}(t)$; because $\mathcal{G} = \mathrm{SL}(2, \mathbb{C})$ is *connected*, the algebraic closure of $\mathbb{C}(t)$ in K is $\mathbb{C}(t)$ itself [16], so $\mathcal{A} \in \mathbb{C}(t)$ is fixed by the Galois action. Now act with explicit elements of $\mathcal{G}$ in the basis (w_1, w_2) . The torus elements $\sigma_\lambda : (w_1, w_2) \mapsto (\lambda w_1, \lambda^{-1} w_2)$ fix $\mathcal{L}$ and scale $W \mapsto \lambda^2 W$, so applying σ_λ to (18) forces $\lambda^{n(k-2)} = 1$ for every $\lambda \in \mathbb{C}^\times$: hence $k = 2$. (In particular $k = 0$ is impossible: $y_0 v$ is not itself algebraic over $\mathbb{C}(t)$.) For $k = 2$, (18) reads $(\rho_0 + \rho_1 \mathcal{L})^n = \mathcal{A} \in \mathbb{C}(t)$, and the unipotent elements $\sigma_\mu : (w_1, w_2) \mapsto (w_1 + \mu w_2, w_2)$ give $(\rho_0 + \rho_1 \sigma_\mu(\mathcal{L}))^n = \mathcal{A}$ with $\sigma_\mu(\mathcal{L}) = 2(w_1 + \mu w_2)'/(w_1 + \mu w_2)$. These are pairwise distinct: equal logarithmic derivatives would force the ratio of $w_1 + \mu w_2$ and $w_1 + \mu' w_2$ to be a constant of K (the constants of K are $\mathbb{C}$), impossible for $\mu \neq \mu'$ since w_1, w_2 are linearly independent over $\mathbb{C}$. Since $\rho_1 \neq 0$, the field elements $\rho_0 + \rho_1 \sigma_\mu(\mathcal{L})$, $\mu \in \mathbb{C}$, are infinitely many distinct roots in K of the polynomial $X^n - \mathcal{A}$ —a contradiction. Hence no such F, a exist.

The quasimodular depth of y_0 is therefore intrinsic—no algebraic factor, at any weight or level, can absorb

the W' term, and $\rho_1 \neq 0$ is exactly the obstruction. This sharpens consequence (ii) of Sec. VI to a categorical exclusion. Both computational inputs (the nonvanishing (17), and, as corroboration, a certified scan excluding any polynomial relation $P(t, tW'/W) = 0$ through bidegree (28, 12)) are certified in `numerics/certify_y0_lemma.py` (`CERTIFICATE_y0_lemma.txt`). (iii) *Physical and arithmetic fingerprints.* The $-\frac{1}{2}$ band-edge exponent above is the square-root van Hove singularity of the three-dimensional density of states. The half-integer residues $\frac{1}{2}$ of V_2 's first-order coefficient a_1 at $t = \frac{1}{4}, \frac{1}{5}, \frac{1}{9}$ are the twist multipliers that defeat every trivial-character ansatz, and the quadratic $15t^2 + 17t - 8$ governing ρ_1 is the one already visible in the rational solution R of Eq. (6). The full symbolic-plus-series certification is the script

`numerics/certify_y0.py` (`CERTIFICATE_y0.txt`).

VIII. THE WATSON REDUCTION: A GENERALIZED BODY-CENTRED-CUBIC INTEGRAL

The orthogonal structure of Sec. V has a concrete origin, which can be exhibited directly. Writing CT_θ for the constant term (torus average) over $\theta = (\theta_1, \theta_2, \theta_3) \in [0, 2\pi)^3$, and eliminating the band index of the dispersive bands from the K_4 (srs) Bloch matrix by a resultant, the moment generating function collapses to a single three-dimensional torus integral,

$$\Phi(t) = \frac{2}{3} - \frac{t}{3} \frac{d}{dt} CT_\theta \log D_+(t, \theta), \quad D_+ = (1 - 6t + 3t^2) - 2t^2 \sum_i \cos 2\theta_i - 8t^{3/2} \prod_i \cos \theta_i. \quad (19)$$

The right-hand side of (19) is exactly the structure function of the *body-centred-cubic lattice with first- and second-neighbour hopping*: $8 \prod_i \cos \theta_i$ is the bcc nearest-neighbour structure function (weight $t^{3/2}$), $2 \sum_i \cos 2\theta_i$ is the simple-cubic second-neighbour shell (weight t^2), and the spectral parameter is $w(t) = 1 - 6t + 3t^2$. The hyperkagome LGF is thus expressible through the classical two-parameter “generalized Watson integral” $I(w; a, b) = CT_\theta [w - a 8 \prod_i \cos \theta_i - b 2 \sum_i \cos 2\theta_i]^{-1}$ and its parameter derivatives, evaluated along the algebraic curve $(w, a, b) = (1 - 6t + 3t^2, t^{3/2}, t^2)$.

Every step is verified in exact arithmetic: the difference-of-squares identity $\det(I - tS^2) = (1 - 6t + (3 - \Lambda)t^2)^2 - t^3 \Xi^2$ for the Bloch matrix S ; the 4:1 measure-preserving torus covering $\kappa = B\theta$ ($\det B = 4$) under which the two invariants become $\Lambda \mapsto 2 \sum_i \cos 2\theta_i$ and $\Xi \mapsto 8 \prod_i \cos \theta_i$; the cancellation of all half-integer powers of t in $CT_\theta \log D_+$ (so Φ is an honest power series in t); and the reproduction of the exact moments ν_m through $m = 8$. Two internal consistencies confirm the picture. The half-integer coupling $t^{3/2}$ is single-valued only on the $\sqrt{t}$ double cover—precisely the determinant-character curve $v^2 = (1 - 4t)(1 - 5t)(1 - 9t)$ of Sec. V. And the singular set $\{\frac{1}{9}, \frac{1}{5}, \frac{1}{4}, 1\}$ is the set of critical values of the bcc(1, 2) band map along the curve, independently reproducing the Riemann scheme (4).

This realisation anchors the modular closed form of Sec. VI on the lattice side: $I(w; a, b)$ is the cubic lattice Green’s function with second-neighbour interactions, whose natural home is the classical second-neighbour literature (Morita–Horiguchi; Glasser’s extended Watson integrals; the cubic modular transformations of Joyce [6])—while the modular structure itself is *proven* in Sec. VI, independently of any particular literature evaluation. The remaining task is to specialise such an evalu-

ation to the curve above, or to integrate D_+ termwise—it is quadratic in each $\cos \theta_i$, so one angular integration yields an inverse square root and the next an elliptic integral, leaving a one-dimensional algebraic period that the orthogonal structure forces into the $\text{Sym}^2(V_2)$ form. We record the reduction here; matching such a classical evaluation, term by term, against the modular parametrization of Sec. VI and the closed form (13) is left to a companion analysis.³

Conclusion. The hyperkagome LGF is a period of an irreducible third-order Picard–Fuchs operator whose differential Galois group is the full orthogonal group $O(3, \mathbb{C})$, with identity component $SO(3, \mathbb{C}) \cong \text{PSL}(2, \mathbb{C})$. It is non-Liouvillian—no closed form in algebraic or elementary functions exists—but it *is* projectively equivalent, through an order-two differential intertwiner, to the symmetric square of a second-order operator V_2 , and V_2 is proven (Sec. VI) to be the uniformizing operator of the modular curve $X(\Gamma_0(30)^+)$: t is a Hauptmodul, given exactly by the level-30 eta quotient (10)–(11), and the LGF is modular at level $30 = 2 \cdot 3 \cdot 5$. Equivalently (this section) it is the classical generalized bcc(1, 2) Watson integral at spectral parameter $1 - 6t + 3t^2$. The obstruction Varma and Monien encountered is therefore not a genuinely third-order (“ SL_3 -generic”) object beyond elliptic reduction; it is a hidden orthogonal/modular structure, invisible to the exponent test because the equivalence is an operator homomorphism rather than a function multiplier or an algebraic pullback—to our knowledge the

³ This reduction is a recent, fast-moving result of the accompanying closed-form analysis; it is independent of the differential-Galois certificate (and of the Maillard communication) and is re-verified in exact arithmetic in the repository.

first three-dimensional lattice Green’s function shown to require one, and the first lattice Green’s function realised at a modular level with three distinct prime factors, extending the level-2, 3, 4, 6 landscape of the classical lattice and sunrise literature [26, 27]. The closed form whose existence Varma and Monien conjectured is thereby established, and the weight-two period y_0 itself is exhibited explicitly in Sec. VII, as a depth-one quasimodular form on $\Gamma_0(30)^+$ twisted by the determinant character—provably not a modular form times an algebraic function at any weight, in particular not an eta quotient. Every structural statement here is a rigorous consequence of M and of the theorems of Secs. VI and VII; that M is the minimal annihilator of the LGF is established to the guess-and-verify standard of the field, with an unconditional creative-telescoping proof the remaining step.

IX. EXACT RESULTS

Independently of the closed-form question, the operator yields exact data. At the symmetric centre $E = 1$ the LGF takes the exact value

$$\operatorname{Re} G(1) = \frac{1}{9}, \quad (20)$$

the flat-band contribution $\frac{1}{3}/(1+2) = \frac{1}{9}$ plus a dispersive principal value that vanishes by the $E = 1$ symmetry. (The value at $E = 2$ is a singular boundary value: $E = 2$ is the van Hove point $t = 1$ of (4), where $\operatorname{Im} G$ diverges logarithmically.) From the exponent $-\frac{1}{2}$ at $t = \frac{1}{9}$ the band-edge density of states behaves as $\rho(E) \sim \sqrt{E_{\text{edge}} - E}$, and the flat band contributes the delta function $\frac{1}{3} \delta(E + 2)$.

X. DISCUSSION

We have given the exact differential equation that controls the hyperkagome lattice Green’s function, determined its differential-Galois structure, proven its modularity, and obtained its weight-two period in explicit closed form. Constructively, M is an irreducible third-order Picard–Fuchs operator that is not a *literal* symmetric square, yet whose symmetric square carries a rational solution; its Galois group is therefore the full orthogonal group $O(3, \mathbb{C})$, with identity component $SO(3, \mathbb{C})$, and M is projectively—through an order-two differential intertwiner—the symmetric square of a second-order operator V_2 . The central result of this paper is that V_2 uniformizes the genus-zero modular curve $X(\Gamma_0(30)^+)$, with t the eta-quotient Hauptmodul (11) and the exact Schwarzian identity (12); the projective monodromy of the hyperkagome operator is the arithmetic lattice $\Gamma_0(30)^+$. This settles the integral left open by Varma

and Monien: the reason a naive elliptic reduction was never found is not that the object is genuinely third-order, but that the reduction is hidden behind an operator homomorphism, invisible to the local-exponent tests that detect only function-multiplier and pullback equivalences—and the closed form, once reached, is modular at level 30 rather than elliptic at the classical levels. The Watson reduction of Sec. VIII anchors the same structure on the lattice side: the LGF is exactly the classical generalized body-centred-cubic integral with first- and second-neighbour hopping, so the classical second-neighbour cubic-lattice literature and the conic/`reduceorder` programme of Assis, van Hoeij and Maillard [13] meet the modular parametrization in a single object. The hyperkagome LGF is, to our knowledge, the first three-dimensional lattice Green’s function shown to require such an operator equivalence rather than a literal symmetric-square factorisation, and the first realised at a modular level with three distinct prime factors.

The result is directly relevant to the modelling of $\text{Na}_4\text{Ir}_3\text{O}_8$ and related hyperkagome magnets, where the single-particle LGF enters impurity, disorder, and dynamical-mean-field treatments; it provides the exact band-edge and special-point data and a certified equation for high-precision evaluation. More broadly, it adds an explicit three-dimensional frustrated lattice, with a nontrivial orthogonal Galois group, a level-30 modular parametrization, and an explicit depth-one quasimodular closed form for its weight-two period, to the catalogue of Picard–Fuchs operators arising in lattice statistics. An unconditional creative-telescoping proof that M is the minimal annihilator and the analogous Heisenberg/transport Green’s functions are natural next steps.

All lattice constructions, exact moments, the certified operator, and the verification scripts are provided in the Supplemental Material and a public repository.

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